RG KEDIA EXAM KIT

B.Sc. Statistics — Semester I

Paper I: Basic Statistics and Probability

Renumbered syllabus, unit-wise important questions, and question papers

Academic Year: 2025–2026

B.Sc. STATISTICS SEMESTER-I EXAM KIT

Paper I: Basic Statistics and Probability

SECTION 1: SYLLABUS

B.Sc., STATISTICS 1st Year: SEMESTER-I (w.e.f. AY 2025-26)

BASIC STATISTICS AND PROBABILITY

UNIT-I: BASIC STATISTICS

Definitions, Measures, Properties and Importance of Central tendencies, Dispersions, Absolute and Relative measures of dispersions, Central and Non-central moments, Skewness and kurtosis and their inter-relationships and computations to the raw & grouped data and data sets. Usage in the domains of image analysis, pattern recognitions, Preparation of descriptive statistical analysis report based on the above descriptive statistics.

UNIT-II: PROBABILITY

Basic concepts used for defining probability, Mathematical, Statistical and Axiomatic definitions of probability, their merits and demerits. Marginal, Joint and Conditional probabilities and independence of events, Addition & Multiplication theorems for 'n' events, Boole's inequality and Bayes' theorem, Problems on computation of Probability and including the usage of theorems.

UNIT-III: RANDOM VARIABLES

Definition of random variable, discrete and continuous random variables, functions of random variables, probability mass, density and distribution functions with their properties and simple problems and illustrations. Notion of bivariate random variable, bivariate distribution, statements of its properties, Joint, marginal and conditional distributions, Independence of random variables. Transformation of one and two-dimensional random variable(s), simple problems on transformation of the random variable(s).

UNIT-IV: MATHEMATICAL EXPECTATION

Mathematical expectation of a random variable, function of a random variable, Computation of raw and central moments, covariance using mathematical expectation with examples, Addition and multiplication theorems of expectation. Generating function, Definitions of moment generating function (MGF), cumulant generating function (CGF), probability generating function (PGF) and characteristic function (Ch.F), their basic properties, applications and computation of those for simple probability functions, Moment inequalities: Chebychev's and Cauchy-Schwartz's inequalities and their applications.

SECTION 2: UNIT-WISE IMPORTANT QUESTIONS

UNIT-I: BASIC STATISTICS

PART A: SHORT ANSWER QUESTIONS

1. Define Basic Statistics.

2. Distinguish between Primary data and Secondary data.

3. Edit of Primary data.

4. Edit of Secondary Data.

5. What is mean by questionnaire? What are the features good questionnaire?

6. What is Schedule? Explain its merits and demerits.

7. Write the differences between questionnaire and Schedule.

8. Mail questionnaire method.

9. What is meant by classification of data? What are the objectives, importance, uses of classification? What are the types of classification?

10. Explain the differences between Classification and Tabulation.

11. Explain the various parts of a statistical table.

12. Explain the relationship between Mean, Median and Mode.

13. "S.D is best measure of dispersion" Justify.

14. Find the effect of change of origin and scale on Moments.

15. What is skewness? What are the merits and demerits of skewness?

16. Explain the differences between Dispersion and Skewness.

17. Define Kurtosis. Write about Measures of Kurtosis.

18. Limits of Karl Pearson coefficient of skewness.

19. Limits of Bowley's coefficient of skewness.

20. Show that for a frequency distribution, $\beta_2 > 1$.

21. Karl Pearson's Measures of Kurtosis.

22. Kelly's Measure of Kurtosis.

23. Inter-Relationship between Skewness and Kurtosis.

PART B: ESSAY / LONG ANSWER QUESTIONS

1. Define Statistics? What are the characteristics of Statistics?

2. Write the Functions of Statistics.

3. Write about Importance/Scope of Statistics/ Application of Statistics in various fields.

4. Define data, primary data and secondary data. What are the various methods of collection the data? Give their advantages and limitations.

5. What is meant by a statistical table? Point out the objectives of tabulation? Explain essentials (characteristics) of good tabulation?

6. Define Measures of Central Tendency (Average). What are the objectives and characteristics of measures of central tendency?

7. Explain the different types of Averages.

8. What is Dispersion? Explain its properties and Uses (Objects).

9. Explain the different types of Dispersion. (Or) Explain about the measures of dispersion with merits and demerits.

10. Define Moment. Discuss its importance. Explain first four central moments in terms of moments about the origin.

11. Explain the various measures of Skewness.

12. Write a detailed note on the applications of descriptive statistics in image analysis and pattern recognition.

13. Explain about the preparation of descriptive statistical analysis report based on the above descriptive statistics.

UNIT-II: PROBABILITY

PART A: SHORT ANSWER QUESTIONS

1. Define Probability.

2. Significance of Probability.

3. Application of Probability.

4. Properties of Probability.

5. Define Event and operations of events.

6. Define trial, Sample Space events.

7. Define Mathematical Statistical and Axiomatic probability.

8. Define Marginal, Joint and Conditional probabilities.

9. Pairwise Independence.

10. Independent events.

11. Mutually Independent Events.

12. Multiplication theorem of probability.

13. Addition theorem of Probability.

PART B: ESSAY / LONG ANSWER QUESTIONS

1. Define the term probability. Explain the basic terminology is used in probability.

2. State the three types of probabilities with an examples.

3. Give the mathematical definition of probability with its merits and demerits.

4. Define probability with statistical and axiomatic approach along with their merits and demerits.

5. Write about the various probabilities under statistical independence.

6. State and prove addition theorem of probability for 'n' events.

7. State and prove multiplication theorem of probability for 'n' events.

8. State and prove Boole's inequality.

9. State and prove that theorem of Baye's.

UNIT-III: RANDOM VARIABLES

PART A: SHORT ANSWER QUESTIONS

1. Define random variable.

2. Define Probability distribution.

3. Define Probability mass function (PMF).

4. Define Probability density function (PDF).

5. Define distribution function.

6. Applications of Probability Distribution.

7. Independence with Multiple RVs.

8. Write short notes on one dimensional transformation of random variables.

9. Transformation of Two-dimensional random variables.

10. The joint pdf $g_{uv}(u, v)$ of the transformed variables U and V is given by $g_{uv}(u, v) = f_{xy}(x, y) \cdot \vert{}J\vert{}$. Where $\vert{}J\vert{}$ is the modulus value of the Jacobian of transformation and $f(x, y)$ is expressed in terms of u and v.

PART B: ESSAY / LONG ANSWER QUESTIONS

1. Define Random Variables. Explain different types of it.

2. Define PMF and PDF. Explain its properties.

3. Explain the concept of distribution function and its properties with suitable examples.

4. Define bivariate random variable. What do you mean by bivariate distribution? Give the statement of its properties.

5. What is transformation of random variable? Explain the transformation of one dimensional random variables.

UNIT-IV: MATHEMATICAL EXPECTATION

PART A: SHORT ANSWER QUESTIONS

1. Define Mathematical expectation of a random variable. Find the expectation of the number on a die when thrown.

2. Write short note on covariance.

3. State and prove addition theorem of expectation for two random variables.

4. State and prove that addition theorem of expectation for 'n' random variables.

5. State and prove multiplication theorem of expectation.

6. What is the effect of change of origin and scale on covariance?

7. Obtain moments through M.G.F.

8. Prove that m.g.f is not independent of changing the origin and scale.

9. Establish the relationship between moments and cumulants.

PART B: ESSAY / LONG ANSWER QUESTIONS

1. Define mathematical expectation and expectation of function of r.v. Derive its important properties with examples.

2. Discuss about the raw and central moments using mathematical expectation.

3. Define variance and derive its properties.

4. Define covariance and also derive its properties.

5. State and prove additional theorem of mathematical expectation for random variables.

6. State and prove multiplication theorem of mathematical expectation for random variables.

7. Define moment generating function (MGF). Discuss its properties, applications and limitations.

8. Define Cumulant generating function (CGF). Discuss its properties and applications.

9. Define probability generating function (PGF). Discuss its properties and applications. Find the relation between pgf and moment.

10. Define characteristic function (Ch.F). Discuss its properties and applications.

11. State and prove Chebyshev's inequality. Explain its applications.

12. State and prove Cauchy-Schwartz inequality. Discuss its applications.

13. If X and Y are independent continuous r.v's then the pdf of $U = X + Y$ is given by $h(u) = \int_{-\infty}^{\infty} f_X(v) f_Y(u-v) dv$.

SECTION 3: QUESTION PAPERS (In Descending Chronological Order)

1. FACULTY OF SCIENCE

B.A / B.Sc. (CBCS) I Semester Examination, July 2026

Subject: Statistics

Paper I: Basic statistics and probability

Code No: H-1206/BL

Time: 3 Hours | Max. Marks: 80

PART - A

Note: Answer any eight questions. (8 x 4 = 32 Marks)

1. Write the usage of statistics in image analysis.

2. Explain the relative measures of dispersion?

3. Define kurtosis? Give its types and applications.

4. State and prove the multiplication theorem for two events?

5. Explain the conditional probability with suitable examples?

6. Discuss the concept of independence of events with two examples?

7. Distinguish between Discrete and Continuous random variables?

8. Define the Jacobian transformation? Give its applications?

9. Define distribution function and mention its applications?

10. Define MGF & CGF. What is the relationship between them?

11. Define mathematical expectation with suitable examples?

12. Derive expressions for variance using expectation of a random variable.

PART - B

Note: Answer all the questions. (4 x 12 = 48 Marks)

1. a) What are the measures of central tendencies and explain its types? Also, write its merits and demerits?

(OR)

b) Define skewness? Explain Karl Pearson's and Bowley's coefficients of skewness?

2. a) State and prove Boole's inequality for n events?

(OR)

b) State and prove the multiplication theorem of probability for 'n' dependent and independent events?

3. a) Explain the transformation of two-dimensional random variable using the Jacobian of transformations method?

(OR)

b) Explain a conditional distribution and conditional expectations with examples?

4. a) State and prove the Cauchy-Schwarz inequality and explain its statistical applications?

(OR)

b) Define characteristic function (CF) and discuss its properties and applications?

2. MODEL PAPER - III (W.E.F from 2025-2026)

Time: 3 Hours | Max. Marks: 80

SECTION - A (Short Answer Questions)

Answer any EIGHT questions. (8 x 4 = 32 Marks)

1. What is Schedule? Explain its merits and demerits.

2. Limits of Bowley's coefficient of skewness.

3. In a certain distribution the first four moments about mean 5 are 2, 20, 40 and 50. Calculate *beta_1* and *beta_2* and state whether the distribution is leptokurtic or platykurtic.

4. Mutually Independent Events.

5. Define Mathematical Statistical and Axiomatic probability.

6. Find the probability that a leap year selected at random will contain 53 Sundays.

7. Applications of Probability Distribution.

8. Define random variable.

9. Two dice are thrown simultaneously and getting a number less than 3 on a die is termed as success. Obtain the probability distribution of the number of successes.

10. State and prove multiplication theorem of expectation.

11. Prove that m.g.f is not independent of changing the origin and scale.

12. Find the mgf of $y = a + bx$.

SECTION - B (Essay Answer Questions)

Answer ALL the questions. (4 x 12 = 48 Marks)

1. a) Explain about the preparation of descriptive statistical analysis report based on the above descriptive statistics.

(OR)

b) Define Moment. Discuss its importance. Explain first four central moments in terms of moments about the origin.

2. a) Define probability with statistical and axiomatic approach along with their merits and demerits.

(OR)

b) State and prove Boole's inequality.

3. a) Define Random Variables. Explain different types of it.

(OR)

b) What is transformation of random variable? Explain the transformation of one dimensional random variables.

4. a) Define covariance and also derive its properties.

(OR)

b) State and prove multiplication theorem of mathematical expectation for random variables.

3. MODEL PAPER - II (W.E.F from 2025-2026)

Time: 3 Hours | Max. Marks: 80

SECTION - A (Short Answer Questions)

Answer any EIGHT questions. (8 x 4 = 32 Marks)

1. Define Kurtosis. Write about Measures of Kurtosis.

2. Kelly's Measure of Kurtosis.

3. First four moments about mean of a distribution are *0, 2.5, 0.7* and *18.75*. Find coefficient of skewness and kurtosis.

4. Define trial, Sample Space events.

5. Addition theorem of Probability.

6. Show that if A and B are independent, *\bar{A}* and *\bar{B}* are also independent.

7. Define Probability mass function (PMF).

8. Transformation of Two-dimensional random variables.

9. If *f(x) = 2x; 0 < x < 1*, find the probability density function of *Y = 8X^3*.

10. State and prove addition theorem of expectation for two random variables.

11. Obtain moments through M.G.F.

12. A random variable X has mean 10 and variance 16. Find a lower bound for *P(5 < X < 15)*.

SECTION - B (Essay Answer Questions)

Answer ALL the questions. (4 x 12 = 48 Marks)

1. a) What is Dispersion? Explain its properties and Uses (Objects).

(OR)

b) Explain the various measures of Skewness.

2. a) Give the mathematical definition of probability with its merits and demerits.

(OR)

b) State and prove multiplication theorem of probability for 'n' events.

3. a) Define PMF and PDF. Explain its properties.

(OR)

b) Define bivariate random variable. What do you mean by bivariate distribution? Give the statement of its properties.

4. a) State and prove Chebyshev's inequality. Explain its applications.

(OR)

b) Define Cumulant generating function (CGF). Discuss its properties and applications.

4. MODEL PAPER - I (W.E.F from 2025-2026)

Time: 3 Hours | Max. Marks: 80

SECTION - A (Short Answer Questions)

Answer any EIGHT questions. (8 x 4 = 32 Marks)

1. Define Basic Statistics.

2. Explain the differences between Dispersion and Skewness.

3. In a certain distribution the first four moments about the point 5 are *-4, 22, -117* and *560* respectively. Find mean, variance and *beta_1*.

4. Significance of Probability.

5. Define Event and operations of events.

6. In a group there are 3 men and 2 women. Three persons are selected at random from this group. Find the probability that one man and two women or two men and one woman are selected.

7. Define Probability distribution.

8. Define Probability density function (PDF).

9. A random variable x is exponentially distributed with parameter 1. Find the density function of *y* = *sqrt{x}*.

10. Write short note on covariance.

11. Establish the relationship between moments and cumulants.

12. x and y are two independent uniform r.v on *(1, 2)* and *(3, 4)* respectively. Find their first four central moments if a, b, c are constants, find *E(ax + by + c)* and *V(ax + by + c)*.

SECTION - B (Essay Answer Questions)

Answer ALL the questions. (4 x 12 = 48 Marks)

1. a) Define Measures of Central Tendency (Average). What are the objectives and characteristics of measures of central tendency?

(OR)

b) Write a detailed note on the applications of descriptive statistics in image analysis and pattern recognition.

2. a) State the three types of probabilities with an examples.

(OR)

b) State and prove addition theorem of probability for 'n' events.

3. a) Explain the concept of distribution function and its properties with suitable examples.

(OR)

b) The random variable X has an exponential distribution, *f(x) = e^{-x}, 0 \le x < \infty*. Find the density function of the random variable *Y = 3X + 5*.

4. a) Define mathematical expectation and expectation of function of r.v. Derive its important properties with examples.

(OR)

b) State and prove Cauchy-Schwartz inequality. Discuss its applications.

5. FACULTY OF SCIENCE

B.A / B.Sc. (CBCS) II-Semester Examination, June / July 2022

Subject: Statistics (Probability Distributions)

Paper-II

Code No. D-6206

Time: 3 Hours | Max. Marks: 80

PART - A

Note: Answer any eight questions. (8 x 4 = 32 Marks)

1. Write short notes on Bernoulli distribution.

2. State and prove reproductive property of Binomial distribution.

3. Define Poisson distribution and obtain its mean.

4. Obtain the recurrence relation for the probabilities of Negative binomial distribution.

5. State and prove lack of memory property of Geometric distribution.

6. Show that Hyper Geometric distribution tends to Binomial distribution.

7. Derive mean deviation about mean of Rectangular distribution.

8. Derive the odd order moments about mean of Normal distribution.

9. State and prove the reproductive property of Normal distribution.

10. Define Beta distribution of second kind. Find its mean and variance.

11. Define the general and standard Cauchy distribution.

12. Write short notes on Central Limit Theorem.

PART - B

Note: Answer all the questions. (4 x 12 = 48 Marks)

1. a) Define Binomial distribution. Obtain its m.g.f and hence find its mean and variance.

(OR)

b) Obtain the first four moments of Poisson distribution and derive coefficient of Skewness and Kurtosis.

2. a) Stating the assumptions show that Poisson distribution is a limiting case of Negative binomial distribution.

(OR)

b) Define Hyper Geometric distribution. Show that Hyper Geometric distribution tends to Binomial distribution.

3. a) Define Rectangular distribution and hence find moments and Skewness.

(OR)

b) Stating the assumptions show that Normal distribution as limiting case of Binomial distribution.

4. a) Derive the following functions of Exponential distributions:

(i) Moment generating function (m.g.f)

(ii) Cumulant generating function (c.g.f)

(iii) Characteristic function (c.f)

(OR)

b) Define Gamma distribution and obtain its moment generating function and hence find its mean and variance.